Assignment Code: DA-AG-014

Introduction to Tableau and Data Visualization| Assignment

Instructions:

- Carefully read each question. Use **Google Docs**, **Microsoft Word**, or a similar tool to create a document where you type out **each question along with its answer**.
  - For **theoretical questions**, write clear and concise answers.
  - For **practical questions**, complete the tasks in **Tableau Public or Tableau Desktop** and **insert relevant screenshots** of your visualizations, dashboards, or Tableau Prep workflows in your answer document.
- Use the provided dataset:
  🔗 **Dataset Link:Global_sales_dataset**
- Once you've completed all questions, **save your document as a PDF** and **upload it to the LMS or as per your instructor's instructions**.
- **Do not zip or archive the file** before uploading.
- Each question carries **20 marks**.

Total Marks: 200

Question 1 : What is Tableau? Explain its importance in Business Intelligence and how it helps in data-driven decision-making.

Answer :

Question 2 : Explain the role of the following Tableau components:

a) Data Pane
b) Worksheet
c) Dashboard
d) Story

Answer :

Question 3 : What is the difference between **Dimensions** and **Measures** in Tableau? Provide examples of each.

Answer :

Question 4 : Define and explain the purpose of Filters, Parameters, and Sets in Tableau.

Answer :

Question 5 : Create a bar chart showing Gross Sales by Country.

- 🔗 **Dataset Link:Global_sales_dataset**
- Sort the countries in descending order of sales
- **Highlight or annotate the bar** that represents the **maximum** and **minimum** Gross Sales.
- Add data labels and format the chart for presentation.

Answer :

Question 6 : Using Tableau, create a dual-axis chart that displays:

- 🔗 **Dataset Link:Global_sales_dataset**
- Monthly Sales as bars
- Monthly Profit as a line
- Filter the data to include only records from the year 2014
- Ensure both axes are synchronized and properly labeled
- Add an appropriate chart title, and format the chart for clear visual presentation
- Paste a screenshot of the final chart in your submission

Answer :

Question 7 : Create a filled map showing total Units Sold by Country.

- 🔗 Dataset Link:Global_sales_dataset
- Add a parameter to allow users to switch between Units Sold and Profit.
- Use the Discount Band as a filter in your visualization.

Answer :

Question 8 : Create a **dashboard** that includes:

- 🔗 **Dataset Link:Global_sales_dataset**
- KPI tiles for **Total Sales**, **Total Profit**, and **Total Units Sold**
- A line chart for **Profit trend over time**
- Filters for **Product** and **Country**

Ensure your dashboard is **interactive and visually appealing**.

Answer :

Question 9 : Your goal is to identify products that generate low profit despite high sales volume.

- 🔗 Dataset Link:Global_sales_dataset
- Use scatter plot or highlight table to identify such products.
- Add filters for Country and Segment.
- Write two business insights based on your chart.

Answer :

Question 10 :

[Scenario-Based – Customer Behavior & Retention Strategy]

Dataset to Use: online_retail_II

Dataset Name: *Online Retail II*
Dataset Source: UCI Machine Learning Repository – Online Retail II Dataset

Business Scenario:

You are a **Data Analyst at an e-commerce company** that sells home decor and gifts across multiple countries. The leadership team is concerned about **customer churn** and **revenue loss** due to inconsistent customer behavior.

They've asked you to investigate patterns in **customer orders**, **returns**, and **geographic sales performance** from the *Online Retail II* dataset.

Your Task in Tableau:

1. **Use Tableau to answer these questions:**
    - Which countries have the **highest number of repeat customers**?
    - What is the **return rate** by product and Find top 10 countries?
    - What time of year do customers tend to buy the most (Seasonality) ?
    - Are there certain customers with **high order value** but also **high return rates**?
2. **Create visualizations:**
    - A map showing **Revenue by Country**
    - A line chart of **Monthly Sales Trend**
    - A bar chart showing **Top 10 customers by Total Revenue**
    - A table/heatmap showing **Top returned products by country**
3. **Build a dashboard for business insights:**
    - Allow filters for **Country**, **Product**, and **Customer ID**
    - Use KPIs for:
        - Total Revenue
        - Total Returns
        - Repeat Customer Count
4. **Write a short business insight (2–3 sentences):**
   Based on your Tableau dashboard, what recommendations would you make to help reduce churn and increase customer loyalty?